TRACK · AGENT-TO-AGENT PAYMENT LOOP

LABLAB.AI · AGENTIC ECONOMY ON ARC · 2026-04

AlphaLoop

Only Track 2 entry fed by a live v1.3 production arb bot. Five specialist agents pay each other sub-cent USDC, amounts varying by signal quality — Arc's USDC-as-gas closes the loop.

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(repo retains legacy "signal-mesh-arc" URL — project renamed to AlphaLoop 2026-04-24)

USDC is the gas.

On Arc.

The hidden human in every "agent economy".

Same story, two different currencies on the ledger — until Arc.

OTHER CHAINS

USDC earned — agent does work

↓

ETH / SOL / MATIC paid in gas

↓

A human has to keep refilling the gas wallet

This is not an agent economy. It's a human economy with an LLM on top.

ARC

USDC earned — agent does work

↓

USDC paid in gas

↓

USDC received by the next agent

Loop closes. No human in the middle.

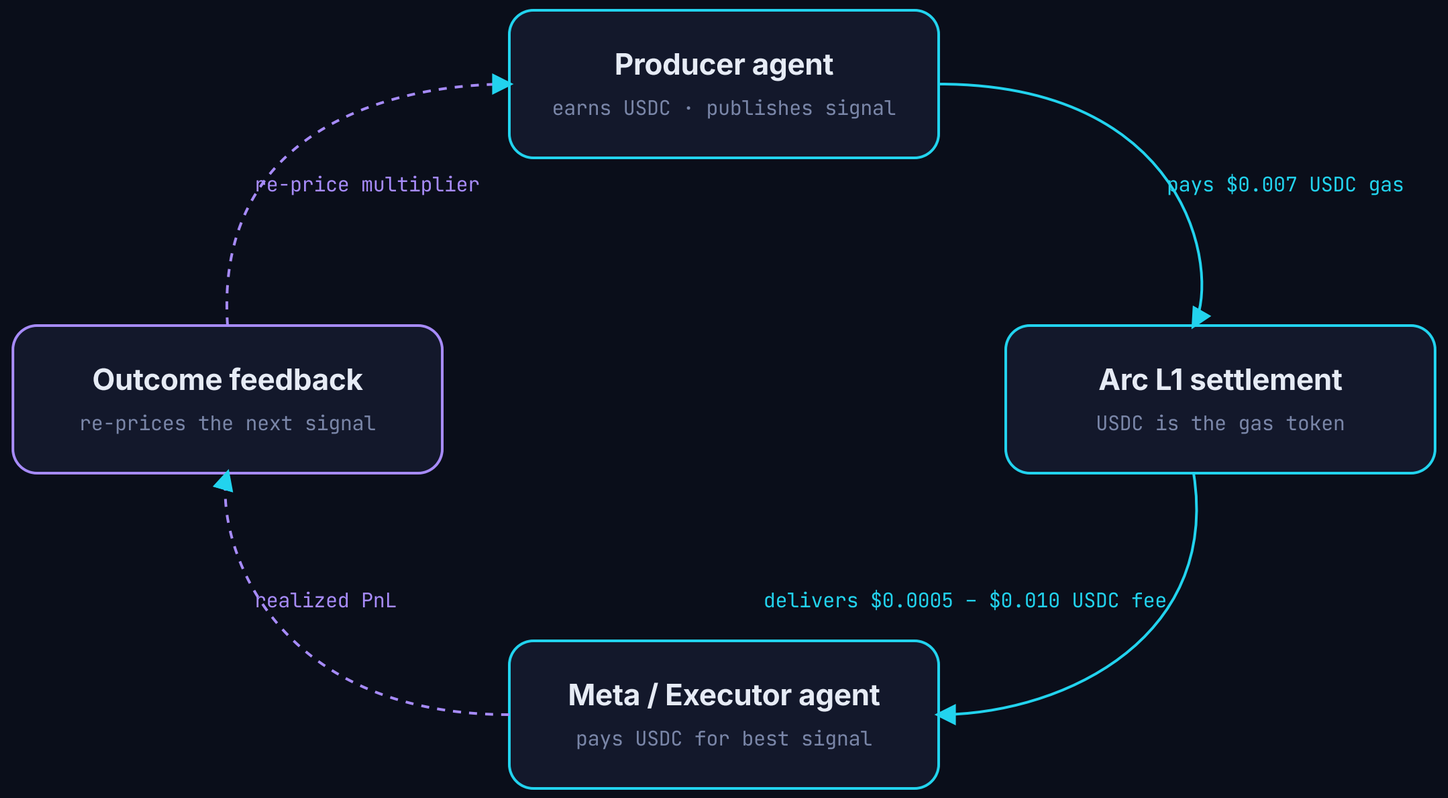
MARGIN COUNTERFACTUAL — SAME $0.005 SIGNAL FEE, THREE CHAINS

Ethereum mainnet	gas ≈ $0.50 per tx → $0.005 fee earns	−$0.495	✗
Polygon PoS	gas ≈ $0.005 → $0.005 fee earns	$0.000	!
Arc (USDC-as-gas)	$0.007 gas + $0.010 fee earns	+$0.003	✓

source: docs/SUBMISSION.md §6 · Arc gas figures from circle_batch_settle.js run receipts

Arc makes USDC the gas token.

Same currency in, same currency out. The loop closes.



Gas and revenue in the same unit → no ETH treasury, no top-up cron, no operator.

Five agents. One bridge. Arc testnet.

Everything below is wired and sending real USDC on-chain.

PRODUCERS · 3

- **Kimchi**
  KRW premium · synthetic AR(1)
- **Dual-Quote**
  Live v1.3 EC2 · 9 coins · pool $1,977
- **Funding**
  Binance fapi · 1,355 real 8h cycles

BRIDGE

Signal registry

- **Variable pricing** — UCB1 bandit picks Q-multiplier ∈ {0.75, 1.0, 1.5, 2.5}
- **Routing** — Gemini 2.5 Flash meta-policy
- **Outcome feedback** — realized PnL → producer

`bridge/src/index.ts · TypeScript`

CONSUMERS · 2

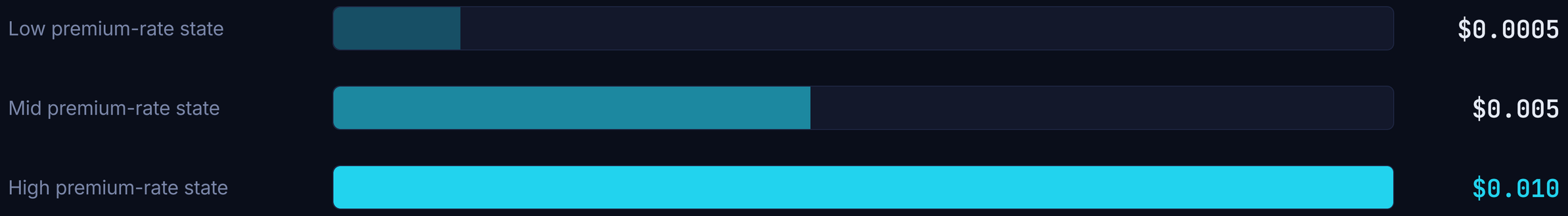
- **Meta agent**
  Gemini 2.5 Flash · local-stub fallback
- **Executor**
  Paper trade + settle USDC on Arc
- **Arc L1 testnet**
  `150 tx + 5 ERC-8004 · chainId 5042002`

Five agents already sending real USDC.

Producers compete · Meta arbitrates · Executor pays — all on Arc, all variable-priced.

The USDC amount on-chain IS the quote.

Not a receipt constant. A live signal-quality score.



price = **fee_base** × Q_{mult} → clipped to [$0.0005, $0.010]

fee_base = FEE_ROUND_TRIP × notional_usd · $Q_{mult} \in \{0.75, 1.0, 1.5, 2.5\}$ via UCB1 over premium-rate state

A producer that keeps being right gets paid more. A producer that keeps being wrong gets paid less. The Q-table sets the multiplier.

Why three strategies, not one?

Each lane has a regime where it shines. The Q-learner discovered the mapping from realized PnL — no hand-coded if-else.

HOT-VOL · COLD-FUND · TIGHT

KR-FOMO regime

KR-Global divergence amplified by vol spike. State 4 — strongest learning signal in the entire Q-table.

Q-VALUE **3.22** runner-up 1.81

v1 · KIMCHI

CALM · HOT-FUND · EITHER SPREAD

Microstructure regime

Funding active, vol calm. Thin, fast cross-venue spreads dominate. States 2 / 3 / 7.

Q-VALUE **0.42** runner-up DUAL_FUND

v2 · DUAL-QUOTE

CALM · COLD-FUND · TIGHT OR WIDE

Carry regime

Pure 8-hour funding-rate carry. Slow, predictable, exogenous. States 0 / 1.

Q-VALUE **0.50** runner-up ALL_V2

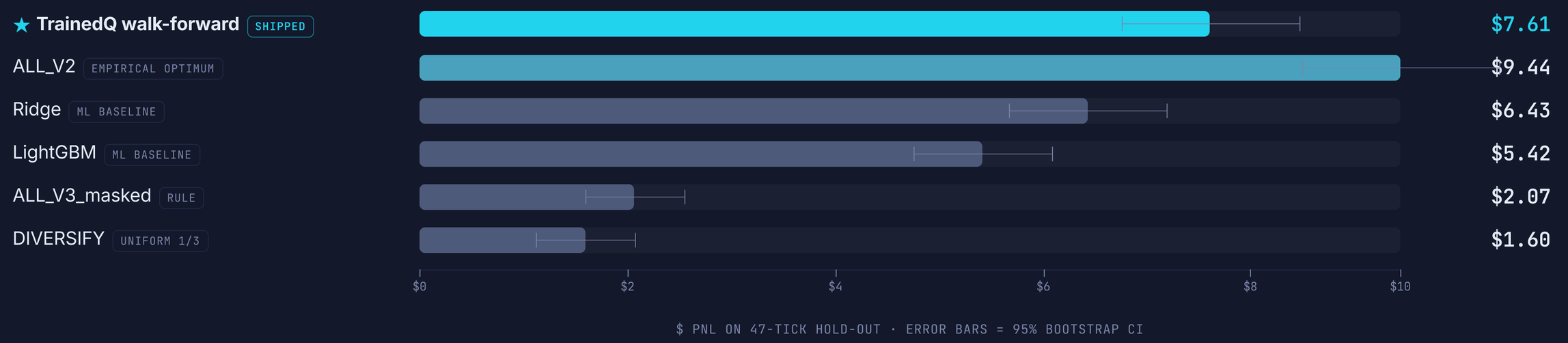
v3 · FUNDING

No human switches fast enough between these as regimes drift.

We let RL learn the switch — from realized PnL, byte-verifiable in `allocator_q.json`.

A blind agent matched the oracle that cheated.

47-tick out-of-sample hold-out · split `cal:178` / `train:46` / `test:47` · calibration frozen before training opens.



- ✓ Statistically ties oracle `p = 0.49 vs ALL_V2`
- ✓ Beats diversify `p = 0.012 vs DIVERSIFY`

Round 1 reward-hacked itself.

Round 2 ties the oracle. A Sutton-school adversarial review caught the bug. Details: `docs/SUBMISSION.md §11.b.1`.

Why p = 0.49 isn't a problem.

The 47-tick hold-out happened to be v2-favorable. In `hot-vol · cold-fund · tight` regimes (state 4), `ALL_V1` q-value is **3.22** vs runner-up **1.81** — a 78% margin. Regime-conditional gaps are the value, not the average. Q-table byte-verifiable in `allocator_q.json`.

Nothing is hidden. Here is exactly what is real.

Every payload on the dashboard carries its provenance tag. Judges should be able to fact-check.

V1 · KIMCHI

SYNTHETIC

AR(1) · SEED COMMITTED

AR(1) mean-reverting premium series with committed seed.

ρ = 0.62 · σ = 0.0031 · seed=2026_04_18

Real cross-venue Upbit KRW + Binance USDT data wasn't available in time. Clearly labeled synthetic in every payload and on the dashboard.

V2 · DUAL-QUOTE

REAL + BOOTSTRAP

1-DAY TAPE · RESAMPLED

1 day of live v1.3 production 1-second tape. Bootstrap-resampled within-regime to fill 90-day window.

2026-04-19 · 86,400 bars · resample within-regime

The 1-second bars are prices the production bot actually saw. Bootstrap is documented as bootstrap, not fabricated.

V3 · FUNDING

REAL

PUBLIC REST · REPRODUCIBLE

Binance

fapi/v1/fundingRate

public REST · 2026-01-21 → 2026-04-21.

5 symbols × 271 ticks = 1,355 real 8h cycles

No API key required. Anyone can reproduce from scratch.

Backtest convergence reproducible from public data alone.

What we used. What we declare.

Circle products, on-chain identity, and custody — stated plainly.

Arc testnet

150 settlement tx + 5 AgentRegistered + 1 deploy. `chainId 5042002`. USDC as gas.

USDC

Unit of account for fee + payment + gas. Dual decimal (18 native / 6 ERC-20) handled.

Developer-Controlled Wallets

4 Circle-SCA wallets via Wallet Sets; 5th `producer_funding` is operator-controlled pending Circle SCA rotation. Signing via `entitySecretCiphertext`.

Developer API + x402

`/v1/w3s/developer/transactions/transfer` posts 150-tx burst. x402 paywall on `/signals/*`.

ERC-8004 registration-v1 · 5 agent identities, on-chain, source-verified.

`AlphaLoopAgentRegistry` at `0xb276b96f…b7ab` emitted 5 `AgentRegistered` events (3 producers · 1 meta · 1 executor) — each carries the sha256 `contentHash` of the agent card at `/.well-known/agent-card/<role>`. Content-addressed off-chain → on-chain linkage. Register tx `0xc331169b…4a8f06`.

Trust model, declared.

4 Circle-SCA wallets + 5th `producer_funding` operator-controlled (pending Circle SCA rotation). Entity-secret leak / KYC freeze / API rate-limit are the three known risks — full disclosure in `SUBMISSION.md` §5 (trust-model section).

What's new here.

Four things we haven't seen in other agent-economy demos.

UNIQUE IN TRACK 2 · 15 COMPETITORS

ERC-8004 on-chain identity — source verified.

Our `AlphaLoopAgentRegistry` contract at `0xb276b96f…b7ab` on Arc testnet emitted 5 `AgentRegistered` events (3 producers · 1 meta · 1 executor) — each event payload carries the sha256 `contentHash` of the agent card served at `/.well-known/agent-card/<role>`. Content-addressed off-chain → on-chain linkage. Contract source is verified on Arcscan. Register tx `0xc331169b…4a8f06`.

merkle root over 150-tx manifest = 400039d5…605b80

Variable on-chain price

The USDC amount you see transferred IS the price. 150 tx on Arcscan, each different — not receipt constants.

Closed outcome loop

Realized PnL retroactively re-prices future signals from the same producer. Round 1 → Round 2 drama: see slide 8.

Regime-aware learned allocation

9 regime cells × 7 actions Q-table. Hot-vol → v1, hot-funding → v2, calm-cold → v3. Mapping is the model's output, byte-verifiable in `allocator_q.json` — no hand-coded if-else.

Most "agent economy" demos stop at "LLM + wallet." We didn't.

No human in the loop.

That's the unlock.

AlphaLoop

GITHUB

github.com/Leewonwuk/signal-mesh-arc

(repo retains legacy URL — renamed in-docs to AlphaLoop)

LIVE DASHBOARD

signal-mesh.vercel.app

JUDGE SCOREBOARD

signal-mesh.vercel.app/scoreboard.html

PITCH VIDEO

YouTube · unlisted